

Banner appropriate to article type will appear here in typeset article

Strain-coupled one-dimensional turbulence: a single-line closure for rapid distortion, its measured limits, and the consequence for leading-edge noise

Sparsh Sharma¹ and Alan R. Kerstein²

¹German Aerospace Center (DLR), 38108 Braunschweig, Germany

²Consultant, Danville, CA 94526, USA

Corresponding author: Sparsh Sharma, sparsh.sharma@dlr.de

(Received xx; revised xx; accepted xx)

XXXX

Key words: one-dimensional turbulence; leading-edge noise; turbulence modelling

1. Introduction

The distortion of turbulence by a mean velocity gradient is among the most basic problems of turbulence dynamics. When a turbulent flow is strained—accelerated through a contraction, deflected around a bluff body, compressed towards a stagnation region—the mean deformation reorganises the fluctuating field: it amplifies some velocity components and suppresses others, redistributes energy among them through the fluctuating pressure, and rescales the spectral content of the motion. When the strain accumulates in a time short compared with the eddy turnover time, the nonlinear interactions cannot keep pace and the response is governed, to leading order, by the linear dynamics of rapid distortion, first analysed by Batchelor & Proudman (1954) and Townsend (1976), developed for flow around obstacles by Hunt (1973) and Hunt & Graham (1978), and given its modern systematic form by Hunt & Carruthers (1990) and Sagaut & Cambon (2018). The turbulence that emerges from a strained region differs from the turbulence that entered it—in intensity, anisotropy and spectral shape—and that difference is often the quantity of practical interest, from grid turbulence in wind-tunnel contractions (Mills & Corrsin 1959; Tucker & Reynolds 1968; Lee & Reynolds 1985) to the distortion of oncoming turbulence approaching a lifting surface (Hunt & Graham 1978; Goldstein & Atassi 1976).

Capturing this distortion at a cost compatible with parametric study is difficult. Scale-resolving simulation represents it directly but precludes broad sweeps over intensity, length-scale ratio and Reynolds number; linear rapid-distortion theory (RDT) is exact in the rapid limit but neglects, by construction, the inter-scale transfer and the slow return to isotropy that act over any finite strain duration (Sagaut & Cambon 2018); second-moment closures

restore the nonlinear relaxation but discard the spectral information that the distortion redistributes (Launder *et al.* 1975; Pope 2000). Between these sits a gap: a reduced-order model that *evolves* strained turbulence with full scale resolution, retaining the rapid distortion and the nonlinear relaxation together, at negligible cost.

A concrete and demanding instance, and the application motivating this work, is turbulence–aerofoil interaction noise. The prevailing prediction framework (Amiet 1975; Paterson & Amiet 1977; Roger & Moreau 2010) obtains the radiated spectrum as a linear functional of the upwash velocity spectrum of the incoming turbulence—but the spectrum it consumes is prescribed, not computed, usually as an isotropic von Kármán form (von Kármán 1948), whereas the turbulence actually reaching the leading edge is strained turbulence. The importance of this pre-impact distortion for the prediction is by now well established: measurements and scale-resolved simulations show the near-edge spectrum systematically altered (dos Santos *et al.* 2023; Ribeiro *et al.* 2023; Piccolo *et al.* 2024), and corrected predictions have been obtained by prescribing RDT-transformed inflow spectra (de Santana *et al.* 2016; dos Santos *et al.* 2023) or by sampling the spectrum close to the edge (Piccolo *et al.* 2024). What these corrections share is that the distorted spectrum is imposed—from linear theory, or from a measurement that a prediction tool does not have. Synthetic-turbulence generators (Kim & Haeri 2015) share the limitation by construction: they reproduce a prescribed spectrum rather than evolving one. A model that predicts the distorted spectrum, including the nonlinear relaxation that linear transforms omit, would supply an element that is at present missing—here and wherever distorted turbulence feeds a downstream prediction, from contraction design to the turbulence ingested by rotors. The subject of this paper is that distortion problem; the acoustic stage enters only as the established linear functional that consumes the model’s output.

Map-based stochastic one-dimensional models offer the route. The one-dimensional turbulence (ODT) model (Kerstein 1999; Kerstein *et al.* 2001), built on the triplet map of the linear-eddy model (Kerstein 1991), evolves a three-component velocity field along a notional line, resolving the full range of length scales while representing turbulent advection as a stochastic sequence of mapping events; a kernel mechanism redistributes energy among components, emulating pressure scrambling (Wunsch & Kerstein 2001; Kerstein *et al.* 2001). Generalised to variable density (Ashurst & Kerstein 2005), adaptive meshes (Lignell *et al.* 2013; Stephens & Lignell 2021) and cylindrical geometry (Lignell *et al.* 2018), embedded as a subgrid closure (Schmidt *et al.* 2010), and coupled to acoustic analogies for jet noise (Sharma *et al.* 2022; Medina Méndez *et al.* 2023), ODT is an accurate and inexpensive generator of scale-resolved turbulence statistics. But standard ODT cannot represent distortion by a mean strain: its pressure treatment models only the slow, turbulence-driven return to isotropy, not the rapid part driven directly by the mean strain, and the apparent remedy—importing the RDT amplitude equation, whose pressure response is the solenoidal projection in three-dimensional wavenumber space—is unavailable on a line that resolves only the wavenumber along its own coordinate. Reconciling the inherently three-dimensional, non-local pressure physics of rapid distortion with the one-dimensional, local structure of ODT is the central modelling problem addressed here.

the following aspects of model formulation and assessment:

The paper reports six results, each framed so that it can be tested:

- A strain-coupled formulation of ODT (SC-ODT) (§ 2): production enters as an exact continuous forcing, the rapid pressure–strain as an energy-conserving redistribution operator built from an explicit spectral closure, and the wavevector kinematics as a dilatation of the domain. The construction preserves the model’s structural invariants and, under the stated closure, reproduces the rapid-distortion evolution of the component energies with no adjustable constant (§ 3).

- The linear reference appropriate to the model's observables: exact RDT projected onto a line is broadband rather than a rigid spectral translation, so the reference against which any line model must be judged already differs from the model's own kinematics at the linear level (§ 4.2).
- An independent spectral benchmark consisting of a strained-box direct numerical simulation together with exact-RDT companion calculations, reduced to the same one-dimensional observables. A three-way comparison localises the model's remaining deficiency in its wavenumber-uniform rapid kinematics and supplies a distance-from-linear-theory envelope as a function of strain rapidity Sk_t/ε (§ 4.6).
- A treatment of what a single line can determine about the rapid pressure-strain term (a non-local three-dimensional object). The obstruction is stated exactly, sharp bounds are obtained from line data alone, and the axisymmetry hypothesis is measured rather than assumed: it is exact for axisymmetric distortion and carries a quantified error under plane strain (§ 5).
- The measured limits of the model and a mechanism that moves them (§ 6). A transmission diagnostic shows the strain-induced anisotropy crossing two triplet-map generations (a factor of nine in scale) undiminished; a controlled family of interventions rules out acceptance gating and map reweighting as remedies. Hierarchical application of the energy-redistributing kernel to eddy sub-scales supplies scale-local relaxation at the scales of the imbalance. The mechanism's deep-strain fade is traced to its event-locked timing and is removed by an independent, size-capped relaxation clock, leaving a single measured allocation bound.
- The consequence for the motivating application (§ 7): the computed distorted spectrum is supplied to Amiet's leading-edge-noise theory in place of the frozen inflow of standard practice, both as a ratio from which aerofoil geometry and observer cancel and as an absolute prediction at the reference operating point. The calculation quantifies the 5–20 dB term omitted by prescribed-spectrum practice; the assembled chain is compared with four published experiments and lies within 1–2 dB rms where the facilities are cleanest.

Throughout we show where the model agrees with reference data and, where it does not, by how much and for what reason.

The paper falls into three parts:

- The model and what a line can know: § 2 reviews standard ODT and derives the strain-coupled formulation; § 3 verifies it against exact rapid-distortion solutions and validates it at the moment level against the DNS of Lee & Reynolds (1985); § 4 constructs the linear reference, examines the emergent distorted spectrum and its operating point, and carries out the strained-box DNS benchmark; § 5 treats the single-line closure question.
- § 6 measures the model's limits and establishes scale-local relaxation—timed by its own clock—as the mechanism that moves them.
- § 7 applies the computed distortion as a correction to the leading-edge-noise input.

~~Section 8 concludes. Supplementary material contains the validation against the strained-turbulence experiment of Chen et al. (2006) (CMK) and the complete intervention family behind the mechanism ledger.~~

2. Strain-coupled one-dimensional turbulence

The linear rapid-distortion equations, recalled below, contain two distinct ingredients: an evolution of the Fourier *amplitudes*, comprising production and the solenoidal pressure projection, and an evolution of the *wavevectors* that carries the kinematic straining of

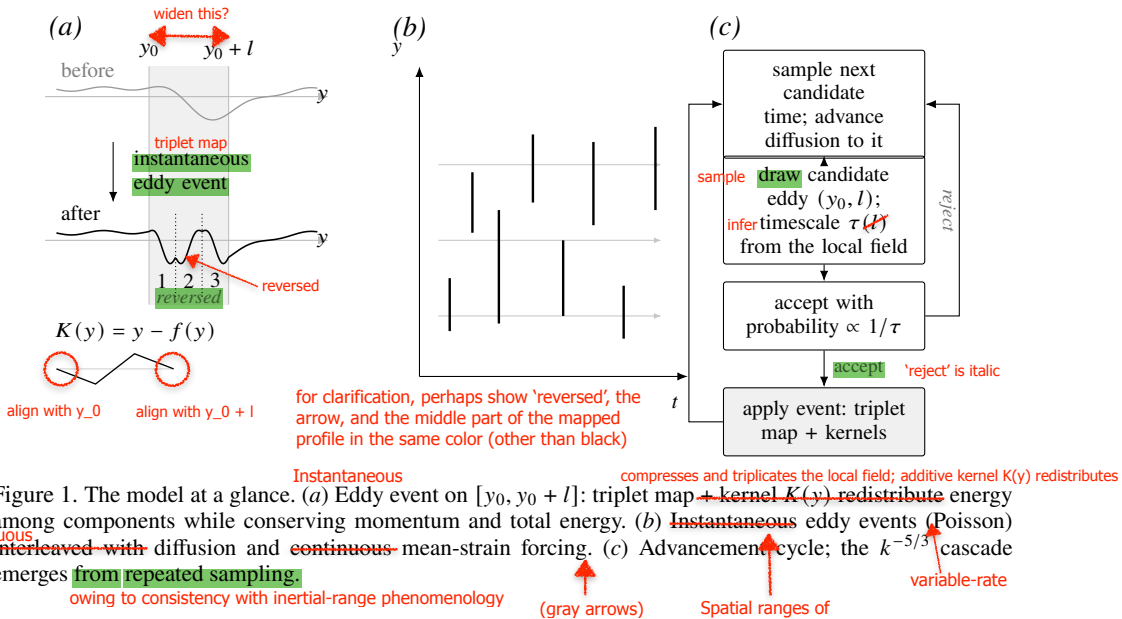


Figure 1. The model at a glance. (a) Eddy event on $[y_0, y_0 + l]$: triplet map + kernel $K(y)$ redistribute energy among components while conserving momentum and total energy. (b) Instantaneous eddy events (Poisson) interleaved with diffusion and continuous mean-strain forcing. (c) Advancement cycle; the $k^{-5/3}$ cascade emerges from repeated sampling.

scales. The pressure part of the amplitude equation requires the full three-dimensional wavevector and is therefore not directly representable on a one-dimensional line; that incompatibility is the focus of this section. The distortion physics is split according to its structure and each part assigned to the mechanism that can carry it faithfully: the production, which is local in physical space, is imposed as a continuous forcing of the line velocity (§ 2.3); the rapid pressure-strain, which is not, is supplied as a continuous, energy-conserving redistribution operator constructed to reproduce the homogeneous rapid-distortion evolution of the component energies under an explicit spectral closure (§ 2.4); and the wavevector kinematics are recovered by a consistent straining of the domain (§ 2.5). The slow, turbulence-driven pressure-strain and the inertial cascade are captured by the discrete eddy events of the standard model, reviewed first.

Throughout, $A_{ij} = \partial U_i / \partial x_j$ denotes the imposed mean velocity gradient, taken uniform across the line at each instant; the medium is incompressible, $A_{ii} = 0$. The line carries the fluctuation field $u_i(y, t)$ about this mean, so that the total velocity is $U_i + u_i$ with $\bar{u}_i = 0$; $R_{ij} \equiv \overline{u_i u_j}$ is the Reynolds-stress tensor, the overline denoting the average over the statistically homogeneous directions (equivalently, the ensemble of independent realisations), and $k_t \equiv \frac{1}{2} R_{ii}$ the turbulent kinetic energy. In the homogeneous configurations used to establish and test the formulation the spatial line average and the ensemble average coincide.

2.1. The standard model

This subsection reviews the one-dimensional turbulence model of Kerstein (1999) in the vector-valued formulation of Kerstein *et al.* (2001), in the notation used throughout; nothing in it is original. The model evolves a three-component velocity field $u_i(y, t)$, $i = 1, 2, 3$, on a one-dimensional domain whose coordinate y is identified with the direction x_2 . The fields represent individual realisations; ensemble statistics are accumulated over many independent simulations. Two interleaved mechanisms advance the state (figure 1): between events the fields obey ordinary one-dimensional diffusion, $(\partial_t - \nu \partial_y^2) u_i = 0$, with kinematic viscosity ν ; and a stochastic sequence of instantaneous eddy events represents the effect of individual turbulent overturns.

Because a continuum rotational motion cannot be represented on a line, an eddy event is an instantaneous, measure-preserving rearrangement of the profiles together with an additive velocity change,

$$u_i(y) \longrightarrow u_i(f(y)) + c_i K(y), \quad (2.1)$$

where f is the *triplet map* (Kerstein 1991, 1999): on an interval $[y_0, y_0 + l]$ of size l it compresses the profile to one third of its length, inserts three copies, reverses the central copy, and leaves the exterior unchanged. It is *measure preserving* (the non-local analogue of a divergence-free rearrangement), *continuous*, and *scale local* (it changes gradients by at most an order-unity factor) (Kerstein *et al.* 2001)—the three invariants whose preservation under an imposed mean strain is established in § 2.3. The additive term is built from the *kernel*

$$K(y) \equiv y - f(y), \quad \int K(y) dy = 0, \quad \int K^2(y) dy = \frac{4}{27} l^3, \quad (2.2)$$

non-zero only within the eddy interval; the first identity guarantees that the additive change conserves the momentum of each component, and the second fixes the energetics of the redistribution.

The kernel term is the model surrogate for the pressure-mediated redistribution of energy among velocity components (Wunsch & Kerstein 2001; Kerstein *et al.* 2001). Requiring that an event redistribute kinetic energy among components while conserving the total, with a rule invariant under permutation of the component indices, gives

$$\Delta E_i = \alpha \sum_j T_{ij} Q_j, \quad \mathbf{T} = \frac{1}{2} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \quad Q_i \equiv \frac{27}{8} \rho_0 l u_{i,K}^2, \quad (2.3)$$

where $u_{i,K} \equiv l^{-2} \int u_i(f(y)) K(y) dy$, Q_i is the available kinetic energy of component i within the eddy, ρ_0 the constant density, and $\alpha \in [0, 1]$ the parameter setting the fraction of extractable energy transferred towards isotropy. Solving the per-component energy balance for the amplitudes yields

$$c_i = \frac{27}{4l} \left[-u_{i,K} + \text{sgn}(u_{i,K}) \sqrt{u_{i,K}^2 + \alpha \sum_j T_{ij} u_{j,K}^2} \right], \quad (2.4)$$

real for $0 \leq \alpha \leq 1$. The transfer matrix $\mathbf{T}$ equalises the component energies: the construction is a model of the return-to-isotropy (*slow*) part of the pressure–strain interaction, a correspondence made precise in § 2.2. The standard model contains no counterpart of the rapid, mean-strain-driven part.

The event sequence is a Poisson process whose rate density $\lambda(y_0, l; t)$ is set by the instantaneous flow state through the standard eddy time-scale construction applied to the quantities Q_i , with a rate constant, a viscous cutoff and a large-eddy suppression parameter as the only tunable constants (Kerstein 1999; Stephens & Lignell 2021); the inertial-range cascade and the $k^{-5/3}$ spectrum are *outcomes* of repeated sampling from this rate, not inputs (Kerstein 1999). Finally, one diagnostic caveat of Kerstein *et al.* (2001) is respected throughout: because the line velocity does not itself advect fluid, off-diagonal products of line velocity components are not momentum fluxes; diagonal component spectra and variances are read from the fields, whereas shear stresses are read from eddy-induced transport.

2.2. Rapid and slow pressure–strain, and linear rapid distortion

The Reynolds-stress transport equation for a fluctuating field about a mean with constant gradient A_{ij} reads, in homogeneous turbulence,

$$\frac{dR_{ij}}{dt} = \mathcal{P}_{ij} + \Pi_{ij} - \varepsilon_{ij}, \quad \mathcal{P}_{ij} = -A_{ik}R_{kj} - A_{jk}R_{ki}, \quad \Pi_{ij} = \left\langle \frac{p'}{\rho} (\partial_j u'_i + \partial_i u'_j) \right\rangle, \quad (2.5)$$

with ε_{ij} the dissipation tensor (Pope 2000). The fluctuating pressure satisfies a Poisson equation whose source splits into a part linear in the mean gradient and a part quadratic in the fluctuations (Pope 2000),

$$\frac{1}{\rho} \nabla^2 p' = \underbrace{-2 A_{lm} \partial_l u'_m}_{\text{rapid}} - \underbrace{\partial_l \partial_m (u'_l u'_m - \langle u'_l u'_m \rangle)}_{\text{slow}}, \quad (2.6)$$

and the pressure–strain correlation inherits the decomposition $\Pi_{ij} = \Pi_{ij}^{(r)} + \Pi_{ij}^{(s)}$ into a *rapid* part driven by the mean strain and a *slow* part driven by the turbulence alone. The slow part relaxes the Reynolds stress towards isotropy—the mechanism that the kernel (2.3)–(2.4) reproduces in model form. The rapid part is linear in A_{lm} and is determined by the velocity spectrum tensor $\Phi_{ij}(\kappa)$; the standard model represents neither $\mathcal{P}_{ij}$ nor $\Pi_{ij}^{(r)}$.

When the strain outruns the eddy turnover—the rapid limit—the response is computed exactly by linear theory (Batchelor & Proudman 1954; Hunt 1973; Hunt & Carruthers 1990). Writing the fluctuation as Fourier modes that follow the mean flow, each mode of time-dependent wavevector $\kappa(t)$ and amplitude $\hat{u}_i$ evolves, in the inviscid rapid limit, according to

$$\frac{d\kappa_i}{dt} = -A_{ji} \kappa_j, \quad \frac{d\hat{u}_i}{dt} = \left(\frac{2 \kappa_i \kappa_m}{\kappa^2} - \delta_{im} \right) A_{mn} \hat{u}_n. \quad (2.7)$$

The bracketed operator combines the production $-A_{im}\hat{u}_m$ with the rapid pressure response, the solenoidal projection $P_{ij} = \delta_{ij} - \kappa_i \kappa_j / \kappa^2$ applied to the strained field; the factor of two is fixed by continuity, since only that coefficient makes $d(\kappa_i \hat{u}_i)/dt$ vanish under (2.7). Two features are central to what follows. First, the pressure response is *non-local* and intrinsically *three-dimensional*: it requires all of $\kappa_1, \kappa_2, \kappa_3$, whereas a single line resolves only the wavenumber along its own coordinate, so (2.7) cannot be evaluated on the line as written. Second, the rapid/slow decomposition furnishes the organising principle of the formulation: the slow part is already carried by the kernel, and it remains to supply the production and a rapid redistribution consistent with (2.7), expressed in quantities available on the line.

2.3. The strain-coupled line equation and its structural admissibility

The production term of the fluctuation momentum equation, $-A_{ij}u_j$, is a pointwise linear combination of the local velocity components and is therefore exactly representable on the line. Between eddy events the molecular evolution is replaced by

$$\partial_t u_i = \nu \partial_y^2 u_i - A_{ij} u_j + B_{ij} u_j, \quad (2.8)$$

in which $-A_{ij}u_j$ is the production and $B_{ij}u_j$ is the rapid pressure–strain operator constructed in § 2.4; the eddy events (2.1)–(2.4) are retained without modification, and the advancement is split between the continuous phase (2.8) and the discrete event sequence.

The forcing must not violate the structural properties on which the model rests, and it does not; we record the argument as a consistency check. The measure preservation, scale locality and kernel identities (2.2) are geometric properties of the map f and kernel K

on a fixed domain, and the operator splitting applies the discrete event exactly as in the standard model, so they hold verbatim. Continuity of the fields is preserved because, for a mean gradient uniform along the line, the operator $\mathcal{A}_{ij} \equiv -A_{ij} + B_{ij}$ acts pointwise and the solution of the non-diffusive part of (2.8) over a sub-step is

$$u_i(y, t) = \left[\exp \int_{t_0}^t \mathcal{A} dt' \right]_{ij} u_j(y, t_0), \quad (2.9)$$

a y -independent matrix, which maps continuous profiles to continuous profiles while the molecular term is smoothing. The forcing displaces no fluid along y , so the Lebesgue measure of the domain is untouched; the deliberate straining of the domain is a separate operation (§ 2.5).

The energetic role of the forcing is exact. At fixed domain length the production term evolves the Reynolds stress as

$$\left. \frac{dR_{ij}}{dt} \right|_{\text{prod}} = -A_{ik}R_{kj} - A_{jk}R_{ki} \equiv \mathcal{P}_{ij}, \quad (2.10)$$

exactly the Reynolds-stress production tensor of (2.5): the term carries no model assumption. The kinetic-energy budget of the formulation thus separates into an exact production by the mean strain, energy-conserving redistribution (the kernels), and dissipation by molecular diffusion—the structure of the exact transport equation. The forcing adds the one term the standard model lacks without disturbing the terms it already represents.

2.4. The rapid redistribution operator

Were the continuous evolution to consist of production and diffusion only ($B_{ij} = 0$), the line would reproduce $dR_{ij}/dt = \mathcal{P}_{ij}$, whereas the exact inviscid rapid-distortion balance is $dR_{ij}/dt = \mathcal{P}_{ij} + \Pi_{ij}^{(r)}$. The omission is not small: for turbulence isotropic at the onset of straining the classical result recovered below gives $\Pi_{ij}^{(r)} = -\frac{3}{5}\mathcal{P}_{ij}$, so production acting alone overstates the initial distortion rate by a factor of $\frac{5}{2}$; under sustained strain it also lets a component energy grow without the bound that incompressibility imposes. Representing $\Pi_{ij}^{(r)}$ is necessary.

2.4.1. The exact rapid pressure–strain and the closure problem

The exact rapid term follows from the amplitude equation. Writing (2.7) as $d\hat{u}_i/dt = M_{in}\hat{u}_n$ with $M_{in} = -A_{in} + 2(\kappa_i\kappa_m/\kappa^2)A_{mn}$, and noting that the incompressible wavevector dynamics preserve the measure $d\kappa$ (their phase-space divergence is $-A_{ii} = 0$), the spectrum tensor $\Phi_{ij} = \langle \hat{u}_i^* \hat{u}_j \rangle$ yields

$$\frac{dR_{ij}}{dt} = \int (M_{in}\Phi_{nj} + M_{jn}\Phi_{in}) d\kappa = \mathcal{P}_{ij} + \Pi_{ij}^{(r)}, \quad (2.11)$$

in which the production separates from the pressure contribution,

$$\Pi_{ij}^{(r)} = 2A_{mn}(\mathcal{R}_{ijmn} + \mathcal{R}_{jimn}), \quad \mathcal{R}_{ijmn} \equiv \int \frac{\kappa_i\kappa_m}{\kappa^2} \Phi_{nj}(\kappa) d\kappa. \quad (2.12)$$

The tensor is traceless—by solenoidality $\kappa_n\Phi_{nj} = 0$, the trace integrand contains $\kappa_i\Phi_{ni} = 0$ —so the rapid term redistributes energy among components without changing the total, in keeping with the energy-conserving character of the kernel mechanism. Equation (2.12) also exposes the central difficulty: the rapid term is a functional of the spectrum tensor through the orientation-weighted integral $\mathcal{R}_{ijmn}$, not of the single-point Reynolds stress—the classical closure problem of the rapid pressure–strain, common to every single-point

model (Crow 1968; Launder *et al.* 1975; Pope 2000) and intrinsic to the physics. A line carries R_{ij} and its own one-dimensional spectra, but not the distribution of energy over wavevector *orientation* that $\mathcal{R}_{ijmn}$ requires; a spectral-tensor closure is therefore unavoidable, and we make it explicit. (What a line's data *do* determine about this integral, and how sharply, is the analytical question taken up in § 5.)

2.4.2. Baseline isotropic closure

The minimal closure takes the spectrum tensor instantaneously isotropic, $\Phi_{nj} = \frac{E(\kappa)}{4\pi\kappa^2} (\delta_{nj} - \kappa_n\kappa_j/\kappa^2)$ with $\int E d\kappa = k_t$. With the angular averages $\overline{e_ie_m} = \frac{1}{3}\delta_{im}$ and $\overline{e_ie_me_ne_j} = \frac{1}{15}(\delta_{im}\delta_{nj} + \delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn})$ for $e_i = \kappa_i/\kappa$,

$$\mathcal{R}_{ijmn}^{\text{iso}} = k_t \left[\frac{4}{15}\delta_{im}\delta_{nj} - \frac{1}{15}(\delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn}) \right], \quad (2.13)$$

and substitution into (2.12) gives, with $S_{ij} = \frac{1}{2}(A_{ij} + A_{ji})$ and $A_{nn} = 0$,

$$\Pi_{ij}^{(\text{r}),\text{iso}} = \frac{4}{5}k_t S_{ij} = -\frac{3}{5}\mathcal{P}_{ij} \quad (\text{isotropic } R_{ij}), \quad (2.14)$$

the exact response of initially isotropic turbulence to a weak rapid strain (Crow 1968) and the content of the isotropisation-of-production model with $C_2 = \frac{3}{5}$ (Pope 2000)—obtained here, with no adjustable constant, as the isotropic limit of (2.12). For anisotropic turbulence $\mathcal{R}_{ijmn}$ must be modelled as a tensor function of R_{ij} ; the general representation linear in the anisotropy is that of Launder *et al.* (1975), to which the construction reduces in form. We adopt (2.13) as the baseline and assess the anisotropic correction in § 3.

2.4.3. Realisation on the line

Given the closed rapid term—symmetric and traceless—we require a pointwise operator B_{ij} whose action on the line reproduces it. Demanding that the operator evolve the Reynolds stress by exactly $\Pi_{ij}^{(\text{r})}$ implies

$$(\text{BR} + \text{RB})_{ij} = \Pi_{ij}^{(\text{r})}, \quad (2.15)$$

a continuous Lyapunov equation with, for positive-definite $\mathbf{R}$, a unique symmetric solution $\mathbf{B}$. The solution conserves kinetic energy automatically—contracting (2.15) gives $2B_{ij}R_{ij} = \Pi_{ii}^{(\text{r})} = 0$, so the rate of working of the operator on the line vanishes. In the isotropic case $\mathbf{R} = \frac{2}{3}k_t\mathbf{I}$ it reduces to the closed form

$$B_{ij} = \frac{3}{5}S_{ij}, \quad \mathcal{A}_{ij} = -A_{ij} + \frac{3}{5}S_{ij}. \quad (2.16)$$

B_{ij} is evaluated from the running Reynolds stress and applied continuously, paced by the mean strain, exactly as the production is. This is deliberate: the rapid pressure–strain is by construction the part of the redistribution driven by the mean gradient and therefore acts at the strain rate, in contrast to the slow redistribution that the eddy kernel supplies at the eddy rate. Strain-driven effects (production and rapid pressure–strain) are continuous and enter through $\mathcal{A}_{ij}$; turbulence-driven effects (the inertial cascade and the slow return to isotropy) remain the discrete events.

The sense in which the formulation is rapid-distortion consistent can now be stated precisely, again as a consistency check rather than a theorem. By (2.10) the production term contributes $\mathcal{P}_{ij}$ to dR_{ij}/dt and by construction (2.15) the operator contributes $\Pi_{ij}^{(\text{r})}$; their sum is the exact balance (2.11) for the prescribed closure, with no adjustable constant, recovering (2.14) at the onset of distortion of isotropic turbulence. Two qualifications

delimit this statement: the consistency holds at the level of the *component energies* and is *conditional on the spectral closure*; the isotropic closure is exact only at onset, and its accuracy for anisotropic states is that of the isotropisation-of-production approximation it reproduces. The *spectral* distribution of the distorted energy is not enforced by (2.8): the operator acts uniformly on every resolved wavenumber, and the distribution of energy across scales emerges from its interaction with diffusion, the cascade and the domain straining. It is therefore a genuine prediction, tested in §§ 3–4—where the uniform, wavenumber-blind action of B_{ij} is also identified as the model’s measured spectral deficiency (§ 4.6.1).

2.5. Wavevector kinematics by domain straining

The second ingredient of (2.7), the wavevector evolution, carries the kinematic compression and stretching of scales and is independent of the amplitude evolution: in the method-of-characteristics reading of linear theory, the wavevectors are the characteristics along which the amplitudes evolve. On the line the relevant wavenumber is κ_2 . We restrict attention to a line aligned with a principal axis of the mean strain, so that $A_{12} = A_{32} = 0$ and a material element of the line remains a coordinate line—as holds for the irrotational mean flow of a stagnation region. Then $d\kappa_2/dt = -A_{22}\kappa_2$, and since a material element of length ℓ stretches as $\dot{\ell}/\ell = A_{22}$ while wavenumbers scale as $1/\ell$, a uniform dilatation of the domain reproduces the wavevector kinematics exactly:

$$\frac{1}{L} \frac{dL}{dt} = A_{22}, \quad \implies \quad \kappa_2(t) = \kappa_2(t_0) \exp\left(-\int_{t_0}^t A_{22} dt'\right), \quad (2.17)$$

with L the domain length, implemented within the Lagrangian adaptive-mesh advancement of the model (Lignell *et al.* 2013) by imposing the mean-flow stretching on the mesh.

This step is not redundant with the amplitude operator. The Reynolds stress is an intensive second moment and is invariant under the uniform dilatation: the dilatation relabels the scales that carry the energy without changing the component energies, while $\mathcal{A}_{ij}$ changes the component energies without relabelling scales. The two operations are the physical-space counterparts of the characteristic motion and the amplitude evolution in (2.7), and together they account for the mean straining without double counting. Two limitations are inherent to representing a three-dimensional kinematics on one axis. First, the transverse wavenumbers also strain but are not represented; their effect is subsumed in the spectral closure, which already supplies the orientation information the line lacks. Second, a rotational mean flow, or a line not aligned with a principal axis, tilts the material line out of the coordinate direction and lies outside the formulation.

2.6. Summary of the formulation

The strain-coupled model advances the line field over a sub-step by composing (a) a *continuous strain–diffusion phase on a dilating domain*—integrate $\partial_t u_i = \nu \partial_y^2 u_i + \mathcal{A}_{ij} u_j$ with $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$, the production exact and B_{ij} given by (2.15) from the running Reynolds stress and the spectral closure (baseline (2.16)), while dilating the domain by (2.17)—and (b) *discrete eddy events* sampled from the standard rate distribution: the triplet map and the slow kernel, unchanged. Step (a) carries the mean-strain physics; step (b) carries the turbulence-driven physics; standard ODT is recovered exactly when $A_{ij} = 0$.

The relative importance of the two steps is governed by the ratio of the mean strain rate $S \equiv (2S_{ij}S_{ij})^{1/2}$ to the turbulence relaxation rate,

$$\chi \equiv \frac{S k_t}{\varepsilon}. \quad (2.18)$$

For $\chi \gg 1$ the distortion accumulates faster than the turbulence can respond, the eddy events are negligible over the distortion time, and the formulation reduces to the rapid-distortion limit of step (a), exact for the component energies under the closure. For $\chi \ll 1$ the turbulence relaxes between increments of strain and the slow kernel dominates. The stagnation region of a leading edge is the intermediate regime $\chi = O(1)$, in which neither limit is uniformly valid and both mechanisms contribute; the fidelity of the formulation across this range, and of the emergent distorted spectrum, are assessed against rapid-distortion solutions and scale-resolving data in §§ 3–4.

3. Verification, and moment-level validation

We separate two questions. *Verification* asks whether the formulation and its implementation reproduce a known exact solution in the regime where one exists—the rapid-distortion limit, tested analytically and in the solver below. *Validation* asks whether the model reproduces the behaviour of real strained turbulence outside that limit; the spectral validation, against the strained-box DNS with its exact-RDT companion, is the subject of § 4, and this section closes with the moment-level validation against the strained-turbulence DNS of Lee & Reynolds (1985). Every test in this section operates at the Reynolds-stress level, by design; no spectral claim rests on it, and none is made before the spectral benchmarks of § 4.

One point of identification should be kept in view throughout this section and the next. The model under test is the strain-coupled formulation (SC-ODT); *standard* ODT is its $A_{ij} = 0$ limit (§ 2.6) and, having no mean-strain coupling—neither the production forcing, the rapid pressure–strain operator, nor the domain dilatation—produces none of the strained curves shown here. Every departure from isotropy in what follows is generated by the added strain coupling, not by the eddy dynamics of the standard model, which supplies only the cascade and the slow return to isotropy. Where the figures refer to “SC-ODT” they mean this formulation; “standard ODT” is reserved for the baseline, and the relaxation mechanism of § 6 (“clock-relaxed ODT”) does not enter until Part II.

3.1. Canonical strains and the exact onset rate

Two irrotational homogeneous strains admit exact rapid-distortion solutions and isolate the physics of interest. *Plane strain*,

$$\mathbf{A} = \text{diag}(a, -a, 0), \quad a > 0, \quad (3.1)$$

is the strain of a two-dimensional stagnation region: with x_1 tangential to the surface, x_2 normal to it and x_3 spanwise, the component compressed by the strain is the wall-normal (upwash) component u_2 , whose amplification is the pre-impact distortion relevant to leading-edge noise. *Axisymmetric strain*,

$$\mathbf{A} = \text{diag}(-\tfrac{1}{2}\gamma, -\tfrac{1}{2}\gamma, \gamma), \quad \gamma > 0, \quad (3.2)$$

is the strain of a contraction or a three-dimensional stagnation flow and provides an independent test in which two components are amplified and one suppressed. For both, $\mathbf{A}$ is symmetric, so the principal-axis alignment assumed in § 2.5 holds; the line is taken along x_2 .

For turbulence isotropic at the onset of straining the continuous evolution (2.8) gives, by (2.10) and (2.14),

$$\left. \frac{dR_{ij}}{dt} \right|_0 = \mathcal{P}_{ij} + \Pi_{ij}^{(r),\text{iso}} = \frac{2}{5} \mathcal{P}_{ij} = -\frac{8}{15} k_t S_{ij}, \quad (3.3)$$

the exact rapid-distortion rate for initially isotropic turbulence, with no adjustable constant. This is a closed-form prediction the model must satisfy identically. Evaluated for the two canonical strains it gives, under plane strain, onset rates $\mp \frac{8}{15} k_t a$ for the stretched and compressed components with the neutral component unchanged, and, under axisymmetric strain, $+\frac{4}{15} k_t \gamma$ for each lateral component against $-\frac{8}{15} k_t \gamma$ for the axis. For plane strain the predicted upwash growth rate $+\frac{8}{15} k_t a$ is to be compared with the $\frac{4}{3} k_t a$ that production alone would give—the factor- $\frac{5}{2}$ overstatement anticipated in § 2.4. A model that passes (3.3) for both strains has its production and rapid terms verified jointly; failure would indicate an error in the operator $\mathcal{A}_{ij}$ or its implementation, isolated from any closure or cascade effect.

3.2. Finite total strain, and the necessity of the rapid term

Beyond onset the turbulence becomes anisotropic and the rapid term (2.12) ceases to be a function of the Reynolds stress alone, so the verification becomes a test of the spectral closure rather than of the construction. We integrate the model moment equation along each canonical strain with two closures: the baseline isotropic closure (2.13) (equivalently, isotropisation of production with $C_2 = 3/5$; IP) and the quasi-isotropic model of Launder *et al.* (1975) (LRR),

$$\Pi_{ij}^{(r)} = C_2 k_t S_{ij} + C_3 k_t (b_{ik} S_{jk} + b_{jk} S_{ik} - \frac{2}{3} b_{mn} S_{mn} \delta_{ij}) + C_4 k_t (b_{ik} W_{jk} + b_{jk} W_{ik}), \quad (3.4)$$

with anisotropy $b_{ij} = R_{ij}/2k_t - \frac{1}{3} \delta_{ij}$, mean rotation $W_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$ (zero for the irrotational strains considered), and $C_2 = 4/5$, $C_3 = 7/4$, $C_4 = 131/100$; the value $C_2 = 4/5$ is fixed by requiring that (3.4) reduce at isotropy to the exact $\frac{4}{5} k_t S_{ij}$ of (2.14), so LRR shares the constant-free onset slope. The benchmark is the exact rapid-distortion solution, obtained by integrating the spectral equations (2.7) over a Gauss–Legendre ensemble of initial wavevector directions on the unit sphere: for initially isotropic turbulence the modal dynamics depend only on direction, so the Reynolds stress is the spherical average of the evolved modal covariance. This integration is independent of the moment models and reproduces the onset slope (3.3) to within 10^{-16} , verifying the rapid-distortion machinery itself.

Figure 2 reports the component-energy fractions against total strain. Three observations follow. First, at onset all curves for each component are tangent—the constant-free identity (3.3) is satisfied—and through $e \lesssim 1$ the closures are nearly indistinguishable from the exact solution. Second, as anisotropy accumulates the closures depart, LRR uniformly less than IP: for axisymmetric strain at $e = 4$ the lateral fraction is 0.498 (exact), 0.468 (LRR), 0.450 (IP), and the axial fraction 0.003, 0.063, 0.100. This gap is the spectral-closure error in isolation—the eddy events are inactive in the rapid limit—and it bounds the accuracy attainable from any single rapid operator. Third, for plane strain both closures miss a *qualitative* feature: in the exact solution the neutral spanwise component grows monotonically, overtakes the upwash near $e \approx 3.3$, and is the largest component by $e = 4$ ($u_3^2/2k_t = 0.51$ against 0.47), whereas both closures predict it to decay. This redistribution into the neutral direction is governed by the orientation distribution of the spectral tensor and cannot be recovered from single-point quantities; it is the moment-level signature of precisely the transverse-wavenumber information a single line does not carry. Section 5.6 returns to it with the strained-box DNS in hand and localises the deficiency in the closure’s b -dependent term.

The upwash component, whose spectrum is the input to the leading-edge response, makes the case for the rapid term quantitative (figure 3). Production acting alone drives the

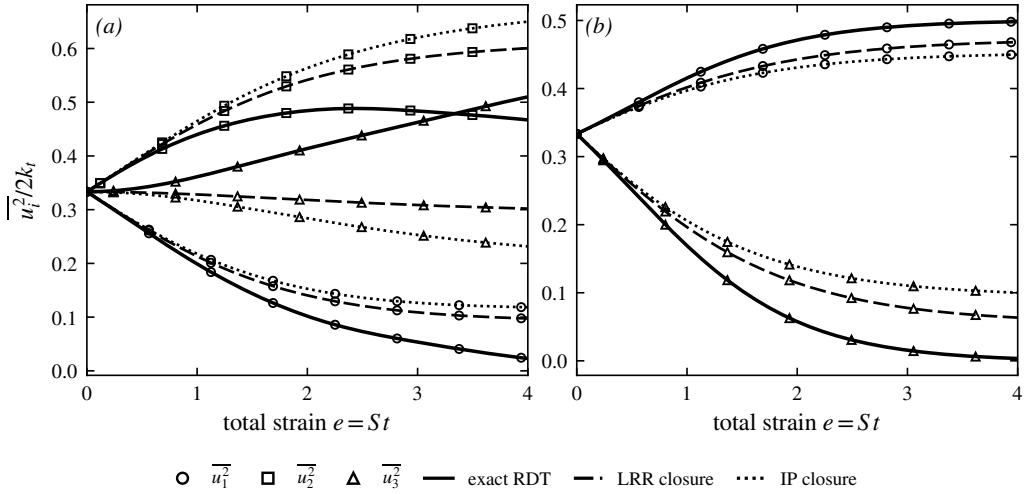


Figure 2. Component energy fractions against total strain for initially isotropic turbulence under (a) plane strain and (b) axisymmetric strain, normalised to $S = 1$: exact rapid-distortion theory (wavevector-ensemble integration of (2.7)) against the SC-ODT moment equation with the LRR closure (3.4) and the IP closure (2.13). Line style denotes the source (solid exact RDT, dashed LRR, dotted IP) and marker the component ($\circ \overline{u_1^2}$, $\square \overline{u_2^2}$, $\triangle \overline{u_3^2}$). For axisymmetric strain the two lateral components coincide and only $\overline{u_1^2} = \overline{u_2^2}$ (lateral) and $\overline{u_3^2}$ (axial) are shown. All closures share the exact onset slope; LRR tracks the exact solution more closely at finite strain.

upwash fraction from $1/3$ to 0.98 by $e = 4$ —almost the entire kinetic energy forced into one component, an essentially non-realisable state—and overstates the initial amplification rate by the factor $\frac{5}{2}$. The rapid term removes three fifths of this at onset and keeps the fraction physical: the exact solution peaks at 0.49 near $e \approx 2.4$ and then relaxes as energy passes to the spanwise component. Both closures reproduce the amplification itself—the effect that the frozen, isotropic input of classical leading-edge-noise prediction omits—while overpredicting it at large strain (LRR 0.60 , IP 0.65 at $e = 4$, against the exact 0.47). The rapid pressure–strain is therefore not a refinement but a necessity, and the accuracy of the method for a given configuration is controlled by the total strain accumulated along the stagnation streamline.

3.3. Verification of the implementation

The preceding checks verify the *formulation*. They do not test the *implementation*, in which the operator is not prescribed but reconstructed at every substep: the line Reynolds stress is formed from the resolved field, the closure supplies $\Pi_{ij}^{(r)}$, the Lyapunov equation (2.15) is solved for B_{ij} , and $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$ is applied pointwise. To isolate this chain the solver is run in the rapid-distortion limit—plane strain, eddy events suppressed, zero molecular diffusion—from a statistically isotropic resolved initial field, and the component energies are evaluated as length-weighted line averages. With the events inactive the line’s component-energy evolution must coincide with the closed moment equation across the whole strain range, not merely at onset.

Figure 4 shows that it does: the solver output lies on the LRR trajectory over $0 \leq e \leq 4$ to within 10^{-4} in each component fraction, so the per-substep stress reconstruction, Lyapunov solve and operator assembly are correct at every state of anisotropy. The comparison must be read correctly: the solver reproduces the *closure*, and it should—with the events and diffusion suppressed the line carries no information beyond the single-point statistics the

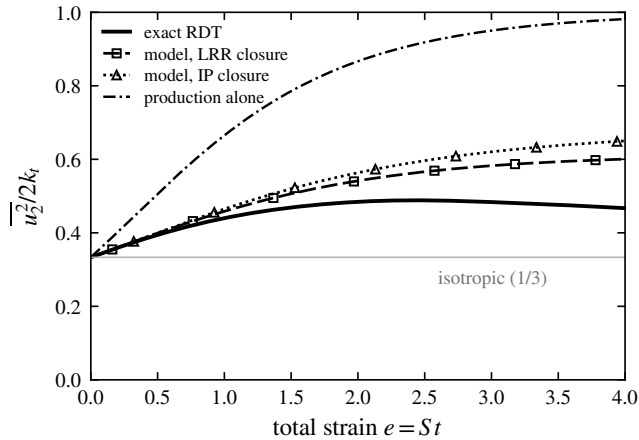


Figure 3. Upwash energy fraction under plane strain for initially isotropic turbulence: exact rapid-distortion theory, SC-ODT with the LRR and IP closures, and production alone ($B_{ij} = 0$).

closure itself uses. The offset from the exact solution is the single-point closure error of § 3.2, not an implementation error; agreement with the exact solution here would indicate a fault. The fractions test only the anisotropy; the kinetic energy, which the traceless rapid term cannot change directly and which grows through the production integral $d \ln k_t / de = f_{22} - f_{11}$, provides the independent magnitude check, and the solver reproduces the LRR amplification to within 0.3%. For definiteness: every strained computation in the remainder of the paper uses the LRR closure, whose error against the exact solution is quantified here at every strain reached; the sensitivity of the spectral conclusions to this closure choice is not estimated but measured, through the exact-RDT comparisons of § 4 and the axisymmetry measurement of § 5, which carry the closure deficiency into the quoted uncertainty rather than leaving it implicit. That assessment requires a resolved spectral initialisation (a white-noise start places energy at the grid scale, where mesh-adaption interpolation damps it) and a strain-resolved timestep bound in the inviscid limit; both effects act nearly isotropically, so the energy panel is what distinguishes them.

Supplementary material generally (but in JFM?) has its own citation and URL; in any case, any mention of something in the supplementary material should be accompanied by the applicable formal designation.

3.4. Moment-level validation against the strained-turbulence DNS of Lee & Reynolds

The DNS of Lee & Reynolds (1985) is retained as a benchmark for one specific reason: it was deliberately run at a turbulence Reynolds number low enough ($q^4/\nu\epsilon \lesssim 100$) to be fully resolvable on a single line, and it reports the Reynolds-stress anisotropy against total strain for several irrotational strain modes alongside the analytic rapid-distortion prediction—a graded, moment-level reference that the spectral benchmarks of § 4 do not provide. The same programme’s later rapid-limit analysis (Lee *et al.* 1986) supplies the analytic reference used in the CMK comparison; higher-Reynolds-number strained computations exist, but the spectral role they would serve is filled here by the dedicated strained-box DNS of § 4, leaving Lee & Reynolds’s low-Reynolds-number study as the appropriate moment-level benchmark for a fully resolved line. It is a Stanford report rather than a journal paper, but it remains the irrotational-strain benchmark of that programme’s own later work: the structure-based models of Reynolds & Kassinos (1995) and Kassinos *et al.* (2001) are validated against its plane-strain and axisymmetric runs (Kassinos & Reynolds 1998). More recent direct simulations of the same deformations at moderate Reynolds number provide independent, journal-published comparators, and are overlaid on the comparisons below: Zusi & Perot (2013) for plane strain (figure 5) and Zusi & Perot (2014) for axisymmetric contraction

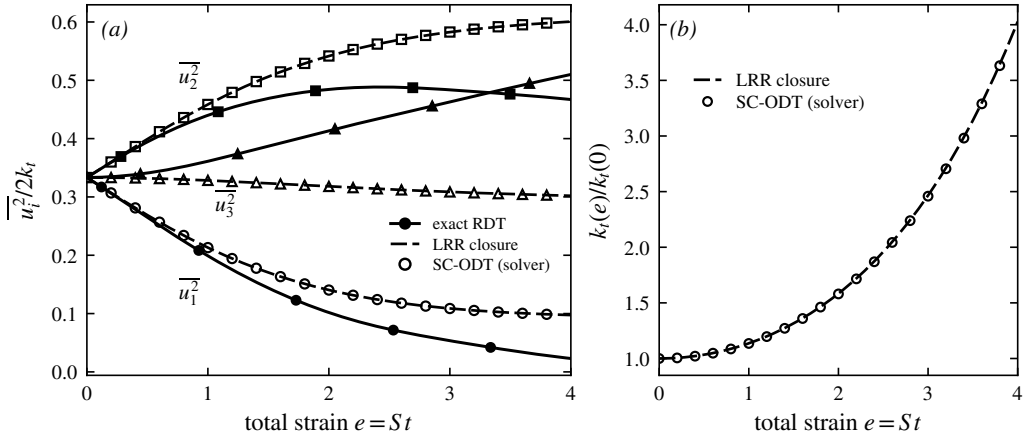


Figure 4. Verification of the SC-ODT implementation under plane strain, eddy events suppressed and zero molecular diffusion. (a) Component energy fractions; (b) kinetic-energy amplification $k_t(e)/k_t(0)$. Open symbols: the SC-ODT line solver (length-weighted line averages), by component ($\circ \overline{u_1^2}$, $\square \overline{u_2^2}$, $\triangle \overline{u_3^2}$); dashed: the LRR closure integrated as a moment equation; solid with filled symbols: exact rapid-distortion theory. The solver reproduces the closure to 10^{-4} in the fractions and 0.3% in the energy; the offset from the exact solution is the closure error of § 3.2, as it must be.

(figure 6), both digitised from their published anisotropy figures and shown over the strain range they reach ($c \leq 1.65$, where $c = \exp \int S dt$ is the deformation ratio, and $\ln c$ is the total strain e). A direct simulation of the CMK straining cycle (Gualtieri & Meneveau 2010), which reports the same variances and pressure–strain budget as that experiment against a Launder–Reece–Rodi closure, is discussed in the supplementary material. The comparison is based on the c dependence of the anisotropy b_{ij} as a function of c against the deformation ratio. computed from the single-point component variances along the line, ensemble-averaged over 1000 realisations; no spectral transform is involved. Two of their strain modes are run: plane strain and axisymmetric contraction, the line along a compressed direction in each case, to the total strains of the DNS ($c = 4$ and 3.32); the measured line compression agrees with $\exp(A_{22}t_f)$, where t_f is the total straining time, to better than 0.3%, confirming the kinematic fidelity of the dilatation. One parameter distinction matters for interpretation: Lee & Reynolds characterise each run by $S^* = Sq^2/\varepsilon$, which equals 2χ identically since $q^2 = 2k_t$ —a correspondence of definitions, not a calibration—and find the axisymmetric anisotropy essentially independent of S^* —a clean, parameter-free target—whereas in plane strain the unstrained component b_{11} is strongly S^* -dependent, spanning $b_{11} \approx 0$ (slow) to $+0.12$ (rapid) in the DNS.

Under plane strain (figure 5) the model reproduces the two strained components quantitatively across the whole range: $b_{22} = +0.21$ at $c = 3.86$ against $\approx +0.20$ in the DNS, and $b_{33} = -0.20$ against ≈ -0.26 , capturing the trend and the bulk of the magnitude. These are exactly the components the DNS identifies as strain-rate-independent, so they constitute the robust part of the comparison. Under axisymmetric contraction the model reproduces the sign structure and the monotonic approach to a strained state while under-predicting the anisotropy magnitude by roughly 25% ($b_{11} = -0.21$ against ≈ -0.29 at $c = 3.16$), dominated by the spectral-closure gap isolated in § 3.2 and common to the whole second-moment family; the line adds only a small artefact, breaking the transverse symmetry with a residual split $b_{22} = +0.102$ against $b_{33} = +0.113$ that the symmetry-respecting mean $\frac{1}{2}(b_{22} + b_{33})$ removes. Figure 6 shows the comparison, with the independent DNS of Zusi & Perot (2014) tracking both the model and Lee & Reynolds over the strain range it reaches.

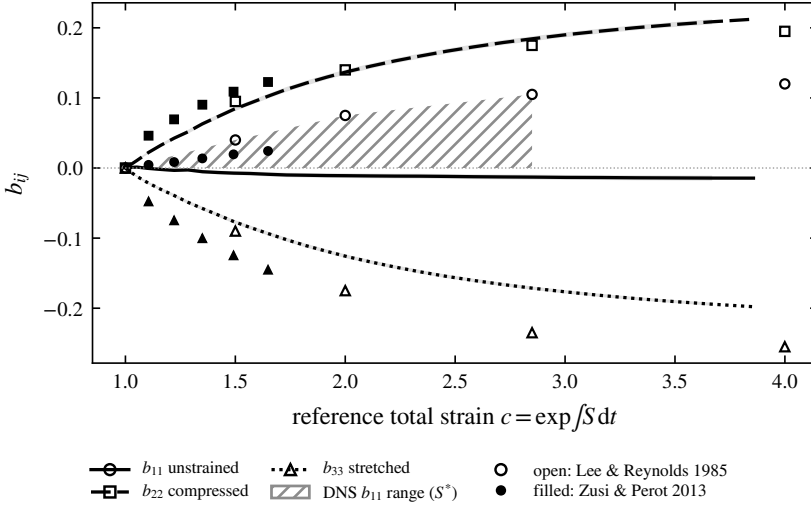


Figure 5. Reynolds-stress anisotropy under plane strain: SC-ODT (lines; ensemble mean over 1000 realisations, standard-error band) against the DNS of Lee & Reynolds (1985) (open symbols, digitised from their figure 5.4). Line style and marker denote the component (b_{11} solid/o, b_{22} dashed/□, b_{33} dotted/△); open markers are Lee & Reynolds (1985), filled markers the independent DNS of Zusi & Perot (2013) (their highest rate, over the strained range $c \leq 1.65$); the hatched band is the DNS b_{11} range over slow-to-fast S^* . The strained components b_{22} and b_{33} are reproduced quantitatively, and the two DNS agree where they overlap; the unstrained b_{11} is strain-rate-dependent in the DNS, with SC-ODT at its slow-strain edge.

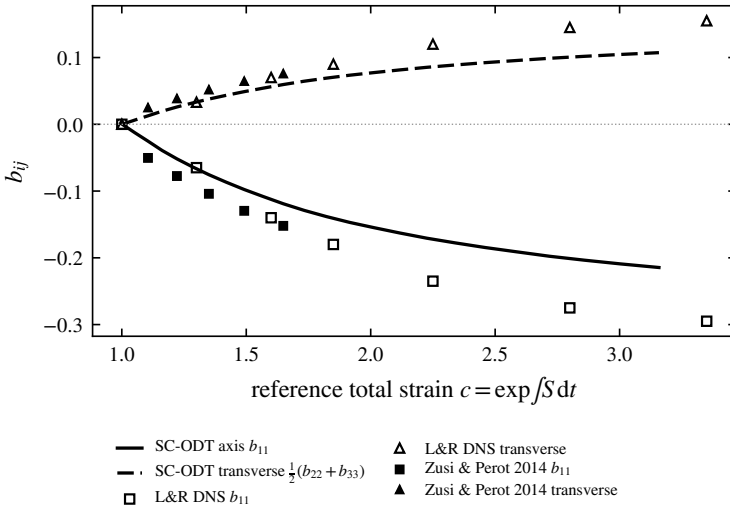


Figure 6. Reynolds-stress anisotropy under axisymmetric contraction: SC-ODT (lines; ensemble mean over 1000 realisations, standard-error band) against the DNS of Lee & Reynolds (1985) (open markers) and the independent DNS of Zusi & Perot (2014) (filled markers, their highest rate, over $c \leq 1.65$). Solid/□: the compressed axial component b_{11} ; dashed/△: the transverse component $\frac{1}{2}(b_{22} + b_{33})$. The model tracks the sign and monotonic growth of both, and the two DNS agree where they overlap.

431 The one component the model does not reproduce, the plane-strain unstrained anisotropy
 432 b_{11} , is not a deficiency specific to the model, and figure 7 locates it exactly. Exact linear
 433 rapid distortion, evaluated by the wavevector-ensemble integration of § 3.2, gives b_{11} rising
 434 to +0.114 at $c = 4$: although $A_{1k} = 0$, the rapid pressure-strain feeds the neutral direction—

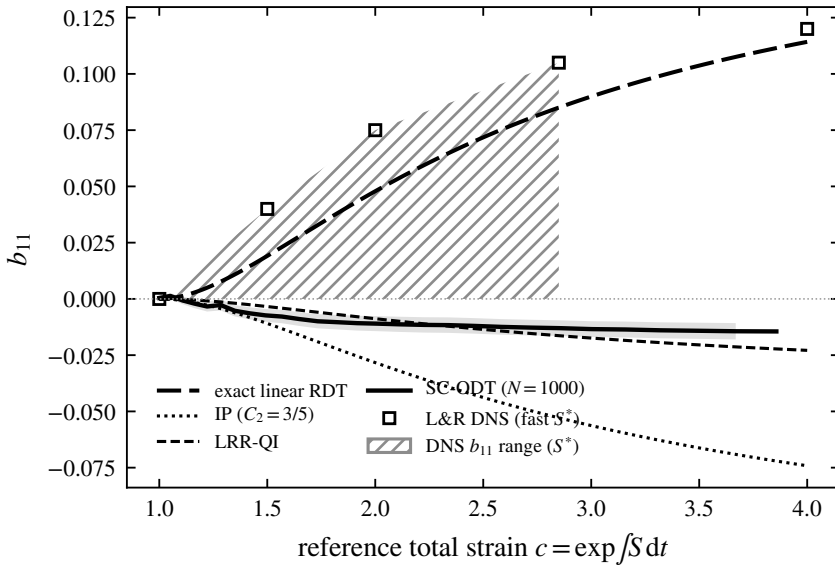


Figure 7. The unstrained-direction anisotropy b_{11} under plane strain: SC-ODT (solid, 1000 realisations, standard-error band) against exact linear rapid distortion (long-dashed, wavevector-ensemble integration), the IP (dotted) and LRR-QI (short-dashed) closures, and the DNS of Lee & Reynolds (1985) (open squares; hatched band = DNS range over slow-to-fast S^*). The fast-strain DNS tracks exact RDT, whose rapid pressure-strain feeds the neutral direction; the single-point closures—and SC-ODT, built on them—miss this orientation-carried redistribution, the deficiency measured and localised in § 5.6.

the same orientation-carried redistribution as the spanwise growth in § 3.2. (A tempting kinematic argument, that $\langle u_1^2 \rangle$ is conserved because the component is unstrained, holds for production but not for the rapid term, and gives the wrong sign; an earlier version of this comparison used it.) The fast-strain DNS therefore *tracks exact RDT*, as Lee & Reynolds themselves observed, while both single-point closures give $b_{11} \approx 0$, slightly negative: the closure family misses the neutral-direction feed, which is carried by the orientation distribution that single-point models discard—the same deficiency measured spectrally, and localised in the closure’s b -dependent term, in § 5.6. SC-ODT coincides with the closure on which its operator is built, so its near-zero b_{11} is correct for its formulation and matches the DNS at slow strain rate, where the nonlinear scrambling erases the neutral-direction feed; at fast strain it inherits the closure family’s known limitation § (3.2) rather than any line-specific artefact. The components that govern the upwash spectrum fed to the leading-edge response are precisely those the model reproduces quantitatively. Finally, Lee & Reynolds’s own transport-budget analysis finds that in plane strain the intercomponent transfer is carried almost entirely by the rapid term—independent support for the rapid/slow decomposition on which § 5 rests.

4. The linear reference and the distorted spectrum

This section establishes what SC-ODT predicts for the strained spectrum and measures it against the correct linear reference and against DNS. It first separates the model’s own kinematics—a rigid spectral translation—from the prediction of linear theory for the line-projected spectrum, which is *not* rigid (§§ 4.1–4.2); it then examines the emergent distorted spectrum and fixes the physical operating point (§§ 4.3–4.4); and it closes with the benchmark that addresses the model-versus-itself comparison—a strained-box DNS with

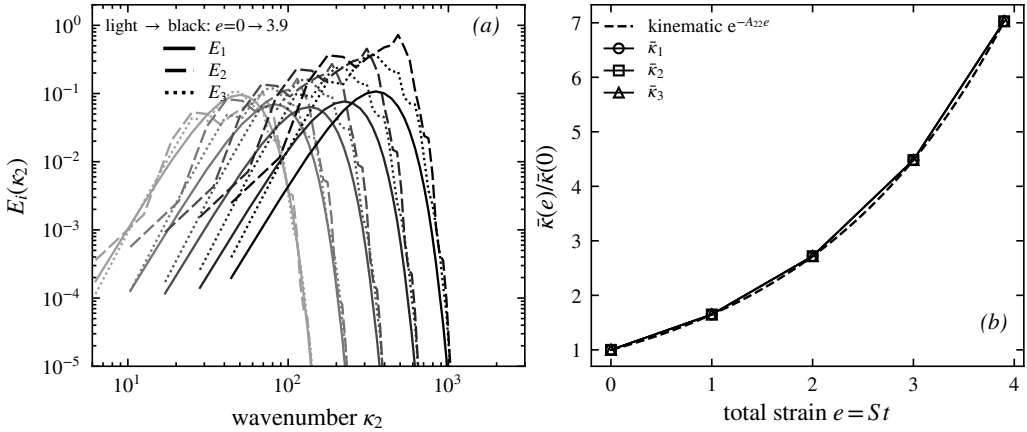


Figure 8. Compression without cascade (eddy events suppressed, inviscid). (a) Component line spectra $E_i(k_2)$ at total strains $e = 0, 1, 2, 3, 3.9$ (lighter to darker); the spectra translate to higher wavenumber without changing shape. (b) Spectral-centroid migration $\bar{\kappa}(e)/\bar{\kappa}(0)$ for each component (symbols) against the kinematic prediction $e^{-A_{22}e}$ (dashed). The centroid follows the prediction and the three components coincide, confirming purely geometric compression of the line.

an exact rapid-distortion companion, reduced to the same one-dimensional observables (§ 4.6).

4.1. Compression without cascade: the model's kinematic reference

With the eddy events suppressed and zero molecular viscosity, the mean strain is the only process acting on the resolved field, and its effect on the line is purely kinematic. The line length contracts as $\mathcal{L}(e) = \mathcal{L}_0 e^{A_{22}e}$, so every resolved wavenumber is rescaled as $k(e) = k(0) e^{-A_{22}e}$ —the wavevector kinematics it shares with rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990)—while the continuous redistribution operator of § 2.4 acts uniformly on every resolved wavenumber. The combination induces that transfer no energy between scales rigid translations of the line spectrum towards higher wavenumber. We emphasise at the outset that this rigid translation is a property of the SC-ODT line—uniform dilatation composed with a wavenumber-uniform operator—and not of linear distortion theory itself; the correct linear reference for the projected spectrum is constructed in § 4.2 below. As a check on the implementation, however, the rigid translation is an exact, parameter-free target.

Figure 8 confirms it. The component spectra translate toward higher k without change of shape—the peak neither broadens nor develops an inertial range—and the spectral centroid follows $e^{-A_{22}e}$, reaching 7.03 at $e = 3.9$ against the exact 7.03, with the three components migrating identically; the domain length follows $e^{A_{22}e}$ to four digits. The single-point moments are unchanged from the no-dilatation case, since a uniform rescaling of position leaves the length-weighted velocity statistics invariant. Compression therefore acts entirely on the spectrum and reproduces the model's kinematic prediction exactly. It is the kinematic reference (the dashed line in all centroid figures below) against which the full model is measured.

4.2. The linear reference for the projected spectrum: exact RDT

The rigid translation above must not be mistaken for the prediction of linear rapid-distortion theory for the quantity the line actually carries. In the full linear theory the amplitude of each Fourier mode evolves at a rate that depends on the orientation of its

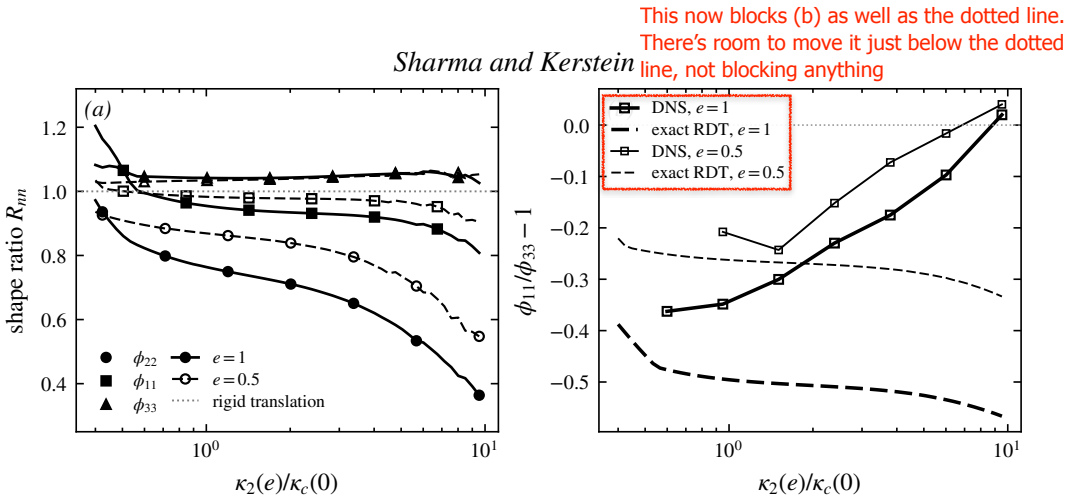


Figure 9. Exact linear theory does not translate the projected spectra rigidly. (a) Shape ratio $R_{nn}(\kappa_2)$ of the exactly distorted, line-projected component spectra to a rigid translation under plane strain, at $e = 1$ (solid, filled markers) and $e = 0.5$ (dashed, open markers); markers distinguish the components, and the dotted line is the rigid translation $R_{nn} = 1$. Closed-form Cauchy solution for an isotropic initial spectrum, integrated over the transverse wavenumber plane. (b) Transverse splitting $\phi_{11}/\phi_{33} - 1$ versus wavenumber in exact linear theory (dashed) and in the strained-box DNS of § 4.6 (solid, squares), at $e = 1$ (heavy) and $e = 0.5$ (light).

three-dimensional wavevector (equation 2.7); the one-dimensional component spectra are projections—integrals over the transverse wavenumber plane—and there is no reason for a projection of orientation-dependent amplification to translate rigidly. Whether it does is a question of calculation, and the answer is that it does not. Using the closed-form Cauchy solution of the linear problem for an isotropic initial spectrum, integrated analytically over the transverse wavenumber plane and band-limited to the wavenumbers resolved by the DNS box of § 4.6, we define the shape ratio $R_{nn}(\kappa_2)$ as the projected component spectrum at strain e , normalised by its integral, divided by the rigidly translated initial spectrum normalised likewise; $R_{nn} \equiv 1$ at every κ_2 if and only if the translation is rigid. At $e = 1$ under plane strain the upwash ratio R_{22} falls monotonically from 0.86 at half the initial spectral centroid to 0.36 at ten times it: exact linear theory redistributes the projected upwash spectrum towards low wavenumber by a factor of 2.4 across the resolved range, because modes aligned with the line ($\kappa \rightarrow \kappa_2$) are amplified least. The transverse components are affected more weakly (R_{11} from 1.07 to 0.81; R_{33} within seven per cent of unity), so the projected anisotropy of linear theory is itself scale-dependent. Figure 9 shows the ratios. The exact-RDT companions of the DNS (§ 4.6), which share its initial fields, reproduce these analytic curves to within 0.05 in the transverse splitting over the resolved band; the analytic integral is used because the companions sample the largest scales with one Fourier line per band.

Two consequences run through the remainder of the paper. First, the comparison of the full model against the rigid translation (§ 4.3) measures the departure of the model from its own kinematics; the departure of the flow from linear theory must be measured against the exact projected reference, and the two differ already at the linear level. Second, the model's rigid spectral kinematics is itself a structural approximation—linear theory deepens the projected anisotropy with wavenumber, which a wavenumber-uniform operator cannot represent—and § 4.6.1 quantifies the resulting deficiency against both the exact reference and DNS.

4.3. The distorted spectrum: strain on against strain off

The central model-internal comparison isolates what the strain does to the spectrum at fixed Reynolds number, holding the turbulence otherwise identical. Two ensembles of 128 realisations are run at $\nu = 10^{-5}$: SC-ODT (eddies and dilatation active) and a baseline identical in every respect except that the strain machinery is switched off, so the line does not compress—which is standard ODT, the $A_{ij} = 0$ limit of § 2.6. Because the two differ only in the presence of the mean strain, the difference between them is the distortion, cleanly separated from the cascade and dissipation the eddies produce on their own.

Figure 10 shows the result, and it departs from the kinematic reference in a specific and physically meaningful way. Consider the centroid first. The no-strain baseline, after the brief initialisation transient, relaxes monotonically to $\bar{\kappa}/\bar{\kappa}_0 < 1$: with no compression, the eddies merely redistribute energy and the centroid drifts below its initial value. SC-ODT is lifted clearly above this baseline—compression does push the spectrum towards higher wavenumber—yet it settles at $\bar{\kappa}/\bar{\kappa}_0 \approx 1.6$ –2.1, far below the rigid-translation line, which rises to 7.4 at $e = 4$. The reason is visible in the spectra: the SC-ODT spectrum is not a rigid translation of the baseline but a *broadband* distortion. It extends to substantially higher wavenumber than the baseline—energy is genuinely carried to small scales by compression and cascade—but the bulk of the energy remains at the energetic scales, and what reaches high wavenumber is dissipated there rather than accumulated.

Two distinct effects are mixed in this departure, and § 4.2 requires that they be separated. The first is the eddy cascade and dissipation, which the strain-on/strain-off construction isolates cleanly at the level of the model. The second is that exact linear theory itself, once projected onto the line, is already broadband (figure 9): part of what a rigid-translation reference attributes to nonlinearity is present in the linear physics. The strain-on/strain-off comparison therefore measures what the mean strain does *within the model*; how the model's distorted spectrum stands against exact linear theory and against DNS on the same footing is answered by the three-way comparison of § 4.6.1. What survives either way is the qualitative point that a frozen, rigidly strained upwash spectrum misrepresents the distorted inflow, which motivates supplying a distorted spectrum to Amiet's theory in place of it, in the spirit of the rapid-distortion-modified inflow spectra of de Santana *et al.* (2016) but without the frozen-shape assumption.

4.4. The physical operating point

The molecular viscosity is, in these units, the inverse integral-scale Reynolds number, and it is a physical property of the represented inflow, not a tunable parameter: a sweep over $\nu = 10^{-4}$ – 10^{-7} shows the cascade deepening monotonically, with the resolved cases settling below the linear-RDT centroid line while the lowest viscosities climb towards it only by piling energy into an under-resolved grid-scale shelf. The operating point is therefore the viscosity matching the target inflow, resolution-checked. For the reference configuration—a NACA0012 at $U_\infty = 20 \text{ m s}^{-1}$ in grid turbulence with intensity $u'/U_\infty \approx 6\%$ and integral scale $\mathcal{L} \approx 55 \text{ mm}$ at the leading-edge plane—the integral-scale Reynolds number is $Re_{\mathcal{L}} \approx 4.4 \times 10^3$, which maps through the initial-condition normalisation to $\nu \approx 3 \times 10^{-6}$, run on a refined grid (16000 cells) on which the dissipation rolloff is resolved with margin; the ensemble comprises 200 realisations.

Figure 11 shows the result. The component spectra fall smoothly across four decades with no grid-scale shelf, the upwash component E_2 amplified above the streamwise E_1 —the spectral signature of the plane strain—and no extended $k^{-5/3}$ range, as expected physically at this Reynolds number (Kolmogorov 1991; Pope 2000). The centroid settles after the initialisation transient and migrates to ≈ 2.4 by $e = 4$, far below the rigid-translation value

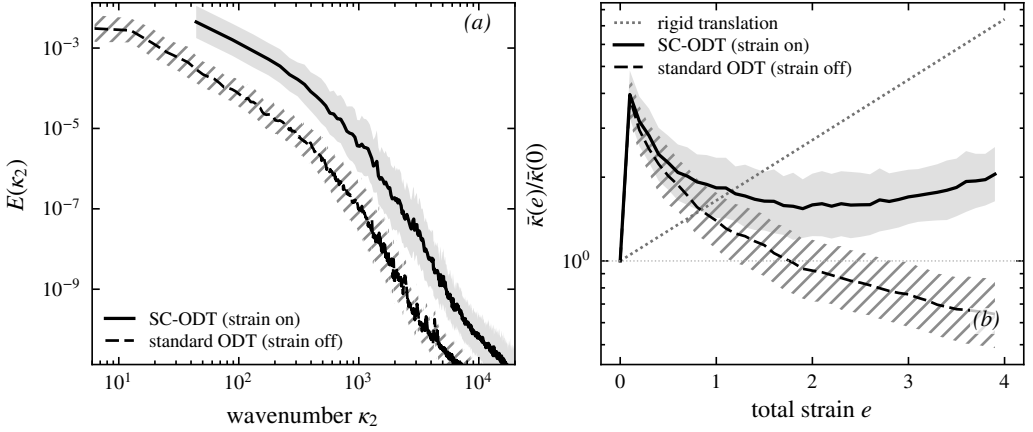


Figure 10. SC-ODT (strain on) against the matched strain-off baseline, standard ODT, at $\nu = 10^{-5}$ (ensemble means over 128 realisations, geometric $\pm\sigma$ bands). (a) Final-strain total spectrum $E(k_2)$: SC-ODT (strain on, solid) extends to higher wavenumber than the baseline (strain off, dashed) but remains broadband, not a rigid translation. (b) Centroid migration: the baseline relaxes below unity, SC-ODT is lifted above it by the compression, yet stays far below the rigid-translation line (grey dotted). The gap between the strain-on and strain-off curves is the distortion attributable to the mean strain at this Reynolds number.

of 7.4: the broadband, dissipative distortion of § 4.3 at the physical operating point. The regime is quantified with an estimator-free dissipation: because the eddy events conserve energy and the production is exact, molecular dissipation is the only sink, so it is fixed by the resolved energy budget

$$\varepsilon = \mathcal{P} - \frac{dk_t}{dt} = -A_{ij}R_{ij} - \frac{dk_t}{dt}, \quad (4.1)$$

every term available on the line. By this budget the strain-rate parameter settles at $\chi = Sk_t/\varepsilon \approx 1.2$ over the strained trajectory (ensemble of 200 realisations), corroborated within a few tens of per cent by the direct gradient estimate $\langle 2\nu(\partial u_i/\partial y)^2 \rangle \approx 1.1$: the operating point sits at $\chi = O(1)$, bracketed by the two strained-box DNS rapidities $Sk_t/\varepsilon = 0.8$ and 16 used in § 4.6.1, the regime the leading-edge stagnation flow occupies, where neither the rapid-distortion nor the near-equilibrium limit applies and where improvement over linear theory is meaningful. The spectral *peak* corroborates the regime reading independently of the dissipation estimator: the energy-containing wavenumber itself migrates far less than the rigid law, so the eddy relaxation holds back the energy-containing range, not merely the dissipation tail—linear theory misplaces the energy-containing wavenumber, not only the spectral shape at high κ . The operating-point ensemble completes in a few hundred core-hours, roughly two orders of magnitude below a strained DNS ensemble at comparable Reynolds number, which permits parametric sweeps over inflow condition.

4.5. Validation against the CMK strained-turbulence experiment

The strain-on/strain-off comparison is model-internal; an external benchmark at laboratory Reynolds number is provided by the CMK plane-strain experiment, whose straining phase (total deformation $D \approx 3$, $e \approx 2.20$ in the present notation) we reproduce. In the rapid-distortion limit (eddy events suppressed, inviscid) the model follows the analytic reference of Lee *et al.* (1986) that CMK use—component centroids on the rigid-translation law with deformation ratio 3.00 at $e = 2.20$, kinetic-energy amplification 1.72 against the exact 1.69, the residual attributed to the moment closure, not the implementation (§ 3.3). With

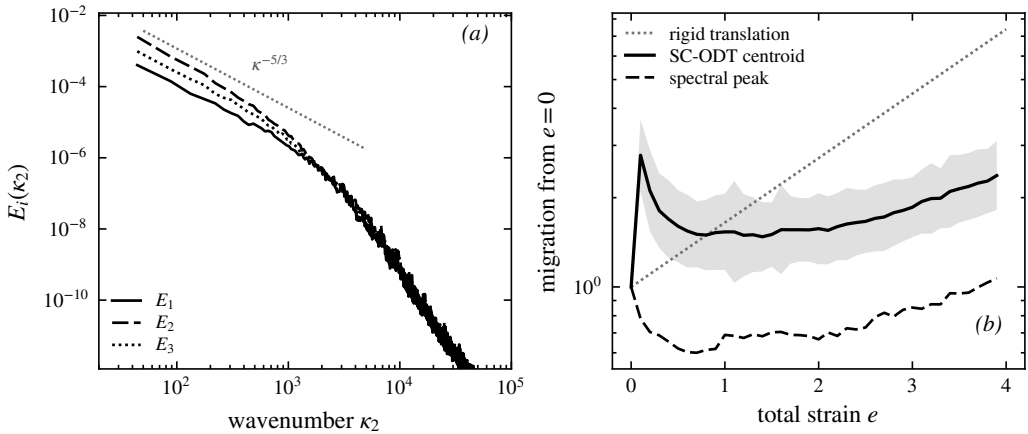


Figure 11. The physical operating point ($\nu \approx 3 \times 10^{-6}$, $Re_{\mathcal{L}} \approx 4.4 \times 10^3$, 16000 cells; ensemble over 200 realisations, geometric $\pm\sigma$). (a) Final-strain component spectra (solid E_1 , dashed E_2 , dotted E_3): smooth rolloff with no grid-scale shelf; the upwash E_2 amplified above the streamwise E_1 ; grey dotted reference slope $\kappa^{-5/3}$. (b) Migration of the total-spectrum centroid (solid) and the spectral peak (dashed) against the rigid-translation law (grey dotted), with the geometric $\pm\sigma$ ensemble spread shaded: the broadband distortion that linear theory cannot represent, at the operating point and with the noise averaged out.

eddies and viscosity active the component centroids diverge under strain—streamwise migration 1.9 against 1.5 for the upwash at the end of the straining phase, both far below the rigid value 3, the upwash held back most—which is the qualitative departure-from-RDT behaviour CMK report and the spectral signature of the broadband distortion of § 4.3. The comparison is qualitative by construction (initial spectrum, spectral projection and scale separation all differ from the experiment’s); the full account, with both figures and the stated limits, is given in the supplementary material (§ S1), and the quantitative spectral assessment is carried out against the strained-box DNS below, where projection and initial spectrum are matched by construction.

4.6. Validation against a strained-box DNS with an exact-RDT companion

The comparisons above are either model-internal (§ 4.3) or, where external, qualitative by construction (§ 4.5) or at the Reynolds-stress level (§ 3.4). The referent quantity for the leading-edge application is spectral, and the objection that the emergent-spectrum results compare the model only with itself must be met with an independent, projection-matched spectral benchmark under the same strain. To that end we performed direct numerical simulations of plane-strained homogeneous turbulence in a deforming-frame (Rogallo-type) pseudo-spectral formulation: a 128^3 box, decaying isotropic precursor, then plane strain $A = S \text{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ applied at strain-rate parameters $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to total strain $e = 1$, four independent realisations ($\kappa_{\max}\eta \approx 0.76$ at $e = 1$: a pilot resolution, so small-scale statements below are qualitative). Alongside every DNS realisation runs an *exact-RDT companion*: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically to the DNS and to ODT. The companion is what § 4.2 used to construct the linear reference; here it completes a three-way comparison in which model, linear theory and DNS are reduced to the same one-dimensional observables. On the ODT side the ensembles use an unstrained precursor: the line is evolved as ordinary ODT until the initialisation transient has relaxed and the eddy population is statistically active, and the strain is switched on only then, so that the strained evolution starts from a physically conditioned state rather

$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c :	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	−0.020	−0.041	−0.050	−0.054	−0.056
DNS	+0.032	−0.033	−0.029	−0.032	−0.036	−0.040	−0.046
SC-ODT	+0.103	+0.103	+0.096	+0.073	+0.075	+0.079	+0.076

Table 1. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. The band below $0.75 \kappa_c$ rests on a few box modes so those results are indicative only.

than from the synthetic initial condition. All ODT statistics in this subsection are medians over 1024 realisations per case with bootstrap confidence intervals; median statistics are used because the strained band energies are strongly intermittent across realisations, and ensemble means of spectral ratios are dominated by rare events at any practical sample size.

4.6.1. Three-way spectral anisotropy: SC-ODT, exact RDT, DNS

The comparison quantity is the strain-induced change of the spectral component anisotropy, $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e , minus the same quantity evaluated in the system's own unstrained state with each mode mapped to its strained wavenumber. Referencing each system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the longitudinal–transverse structure of the isotropic line spectra, which differs between a three-dimensional box and the component-symmetric ODT line. Figure 12 and Table 1 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial spectral centroid.

The structure of the tabulated results is the finding. Exact linear theory concentrates the strain-induced upwash anisotropy at the energy-containing scales and removes it from the small scales, because modes aligned with the line are amplified least; the DNS shows the same shape, weakened at the large scales by the nonlinear return—that nonlinear effects absent from linear theory shape the strained anisotropy across scales is likewise the finding of the 4096³ axisymmetric-contraction simulations of Clay & Yeung (2016). SC-ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the signature, anticipated in § 4.2, of a wavenumber-uniform redistribution operator acting on a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks the closure at the moment level, § 3.4), so the deficiency is one of allocation across scales, not of amount. The transverse splitting $\phi_{11}/\phi_{33} - 1$, a combination of the two transverse components just shown, carries the same reading (not plotted): exact RDT deepens it with wavenumber and the DNS relaxes it towards zero at the small scales, whereas SC-ODT holds it essentially constant—the wavenumber-uniform signature again, now in the neutral-to-streamwise balance rather than the upwash. For the leading-edge application the operative consequence is that the high-wavenumber anisotropy of the upwash spectrum, which the Amiet response weights, has the wrong sign in SC-ODT; the closure element that must carry the correction—a wavenumber-dependent rapid operator in place of the uniform one—is exactly the spectral kernel formulation of § 5, whose introduction ~~exercise~~ is thereby motivated by a measured deficit rather than by its association with an identified physical process.

4.6.2. Distance from linear theory as a function of strain rapidity

The three-way comparison resolves the anisotropy by wavenumber at one strain-rate parameter. A complementary scalar measure tracks the *whole* spectral shape across strain

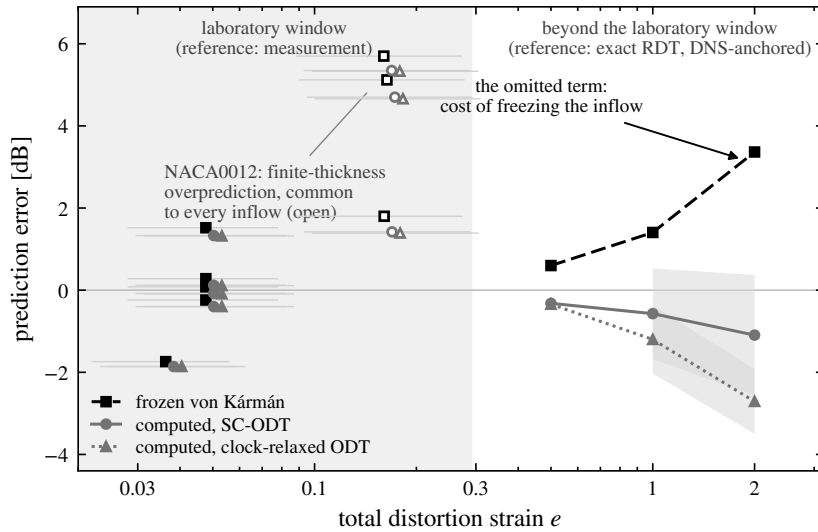


Figure 22. The cost of the frozen-inflow assumption against accumulated distortion: signed far-field prediction error, judged by the best available reference at each strain. Laboratory window (shaded): the eight measured cases of figure 20, bias over each case’s comparison band—flat plates filled, NACA0012 open (its common overprediction is the finite-thickness effect); horizontal bars span the effective-strain truncation range, and the three inflow chains are offset slightly in e for legibility. Beyond the window: band-mean error over 100–1000 Hz against the exact-RDT reference of figure 21 at $e = 0.5, 1$ and 2 ; shaded bands give the $\pm 10 \log_{10}(1 + \text{rms})$ spectral-distance uncertainty where measured ($e = 1, 2$).

prediction but the magnitude and sign of the term that frozen-spectrum practice omits: order 5–20 dB at the energy-containing reduced frequencies for total strains of 0.5–2—far too large to neglect, and of the opposite sign to the blocking correction with which it must ultimately be combined. Figure 19 shows that the chain from model output to absolute far-field level is complete and cheap; what remains for a predictive comparison against measurement is the blocking correction and the streamline strain history, for which the present model supplies the distorted-inflow input at negligible cost, with its uncertainty quantified by the envelope of § 4.6.2 and its fidelity improved by the mechanism of § 6 precisely in the wavenumbers the response weights.

8. Conclusions

IS THERE ANYTHING WORTH MENTIONING HERE ABOUT THE CMK COMPARISON IN SEC. 4.5 AND ITS ELABORATION IN SEC. SM1?

We have presented a strain-coupled formulation of one-dimensional turbulence and, around it, a systematic account of what a single line of turbulence can and cannot say about rapid distortion. The formulation splits the distortion physics by structure: production enters as an exact continuous forcing, the rapid pressure–strain as an energy-conserving operator built from an explicit spectral closure, and the wavevector kinematics as a dilatation of the domain, while the eddy events of the standard model continue to carry the cascade and the slow return to isotropy. The construction preserves the model’s structural invariants, reproduces the constant-free rapid-distortion onset rate identically, and is verified in the solver to the level of the closure it evolves (10^{-4} in the component fractions); at the moment level it reproduces the strained components of the Lee & Reynolds (1985) DNS quantitatively, with the one exception—the neutral-direction anisotropy—traced to the orientation information that every single-point closure discards, not to the line.